Department of Mechanical and Production Engineering

Aarhus University

Exercises

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Course: Dynamik og Vibrationer — 290211U001

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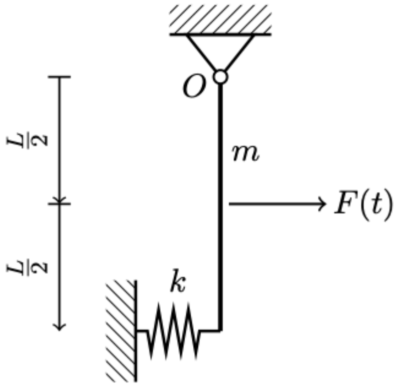
Lecture 1: Introduction

26. August 2025

1.1 Exercise 1.1

We consider the rotation of a rigid body shown in **Figur 0.1** about the point O. The rod has mass m and the spring fastened to the rod at distance L from the point O has stiffness k. A force $F(t)$ is applied at a distance $L/2$ from the point O.



Figur 0.1

a) Derive the non-linear equation of motion for the rotation of the rigid body.

We determine the resultant moment around O as:

$$I_O\ddot{\theta} = -kL\sin\theta L\cos\,\theta - mg\frac{L}{2}\sin\theta + F\frac{L}{2}\cos\theta$$

$$I_O\ddot{\theta} + \sin\theta\left(kL^2\cos\theta + mg\frac{L}{2}\right) = F\frac{L}{2}\cos\theta$$

.

The moment of inertia for a rod rotating about its end is given by $I_O = \frac{1}{3}mL^2$. Therefore we get:

$$\frac{1}{3}mL^2\ddot{\theta} + \sin\theta\left(kL^2\cos\theta + mg\frac{L}{2}\right) = F\frac{L}{2}\cos\theta.$$

b) Linearize the non-linear equation of motion using the assumption of small rotations.

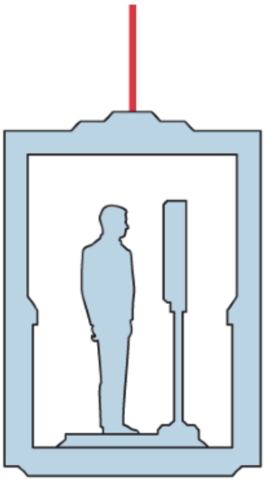
By assuming $\cos\theta \approx 1$ and $\sin\theta \approx \theta$ we get:

$$\frac{1}{3}mL^2\ddot{\theta} + \theta\left(kL^2 + mg\frac{L}{2}\right) = F\frac{L}{2}.$$

1.2 Exercise 1.2

A person weighing 75 kg is standing on a scale in an elevator as shown on **Figur 0.2**. During the first 3 seconds of vertical motion upwards from rest, the force in the cable is 8300 N. Determine what the scale shows (in Newton) in this configuration. It is noted that the total mass of the elevator, person and scale is 750 kg.

Let $m_p = 75\,\text{kg}$ denote the mass of the person and $m = 750\,\text{kg}$ denote the entire mass of the system. Also let



Figur 0.2

$F_c = 8300\,\mathrm{N}$ denote the force in the cable. Newton's second law on the entire system gives

$$m\ddot{d} = F_c - mg \implies \ddot{d} = \frac{F_c - mg}{m}.$$

For the person inside the elevator using Newton's second law gives

$$m_p\ddot{d} = F_R - m_p g \implies F_R = m_p\ddot{d} + m_p g$$

where F_R is the reactant force the scale exerts on the person (i.e. the force the person exerts on the scale). Substituting in the previous expression for $\ddot{d}$ we get

$$\begin{aligned} F_R &= m_p\left(\ddot{d} + g\right) \\ &= m_p\left(\frac{F_c - mg}{m} + g\right) \\ &= m_p\left(\frac{F_c}{m}\right) \\ &= \frac{m_p F_c}{m} \\ &= \frac{75\,\mathrm{kg}\cdot 8300\,\mathrm{N}}{750\,\mathrm{kg}} \\ &= 830\,\mathrm{N}. \end{aligned}$$

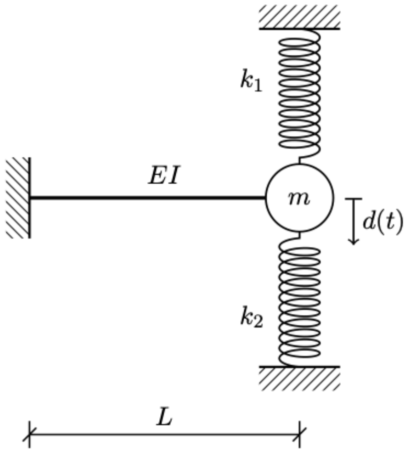
Therefore the scale will show a reading of 830 N.

Lecture 2: Free motion/eigenmotion of SDOF systems

2. September 2025

2.1 Exercise 2.1

We consider the shown system, which is to be analysed as an equivalent single-degree of freedom system (SDOF-system) with the transverse motion of the point mass m as the degree of freedom. It is noted, that the stiffness of the beam (without taking the discrete springs into account) against transverse direction in the free end due to a transverse force acting in this point is $k_{\text{beam}} = \frac{3EI}{L^3}$, where L, I, and E is the length, area-moment of inertia and the elastic modulus of the beam. Assume, that the beam and the discrete springs are mass-less.



a) Determine (symbolically) the undampened angular eigenfrequency ω_u of the equivalent SDOF-system.

The undampened angular eigenfrequency ω_u is given as:

$$\omega_u = \sqrt{\frac{k}{m}}.$$

In this case the three springs are connected in parallel and so:

$$k_{eq} = k_{\text{beam}} + k_1 + k_2 = k_1 + k_2 + \frac{3EI}{L^3}.$$

And therefore the undampened angular eigenfrequency of the system is

$$\omega_u = \sqrt{\frac{k_1 + k_2 + \frac{3EI}{L^3}}{m}}.$$

b) It is now noted that (in SI-units) $EI = 100$, $L = 1{,}3$, $m = 0{,}5$, $k_1 = k_2 = 25$, $d_0 = 0{,}1$ and $\dot{d}_0 = 0$. Determine the undampened eigenperiod of the equivalent SDOF-system.

Note that the connection between frequency as determined in a) and period is:

$$T = \frac{2\pi}{\omega_u} = \frac{2\pi}{\sqrt{\frac{2\cdot 25 + \frac{300}{1{,}3^3}}{0{,}5}}} = 0{,}33\,\text{s}.$$

c) Determine the maximal speed, that the equivalent SDOF-system experiences in its eigenmotion.

The displacement $d(t)$ is given as:

$$d(t) = A\cos\left(\omega_u t - \phi\right).$$

And the speed $\dot{d}(t)$ can be found simply vie differentiation as

$$\dot{d}(t) = -\omega_u A\sin\left(\omega_u t - \phi\right).$$

In its maximum this is

$$\max(\dot{d}(t)) = \omega_u A.$$

We also know that:

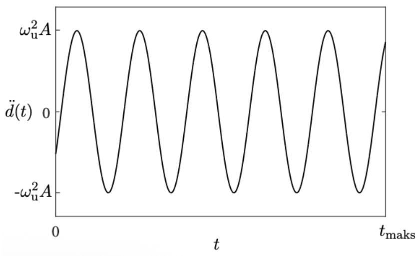
$$A = \sqrt{A_1^2 + A_2^2} = \sqrt{d_0^2 + \frac{\dot{d}_0^2}{\omega_u^2}} = d_0.$$

Therefore the maximum velocity is:

$$\max\left(\dot{d}(t)\right) = \omega_u A = \sqrt{\frac{k_1 + k_2 + \frac{3EI}{L^3}}{m}} \cdot d_0 = 1{,}73\,\frac{\mathrm{m}}{\mathrm{s}}.$$

2.2 Exercise 2.2

With reference to the following harmonic response choose the correct starting conditions.



1. $d_0 < 0 \wedge \dot{d}_0 = 0$
2. $d_0 > 0 \wedge \dot{d}_0 = 0$
3. $d_0 < 0 \wedge \dot{d}_0 > 0$
4. $d_0 > 0 \wedge \dot{d}_0 < 0$
5. $d_0 = 0 \wedge \dot{d}_0 > 0$
6. $d_0 = 0 \wedge \dot{d}_0 < 0$

The plot looks to be showing sinusoidal acceleration. In the initial case the acceleration is negative, this means that the speed here is necessarily also negative, a negative speed means that the initial displacement must have been positive, hence the correct answer is 4

2.3 Exercise 2.3

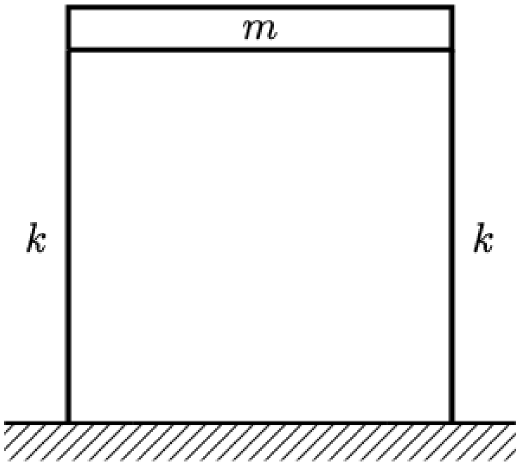
We consider the shown system, which is to be analysed as an equivalent SDOF-system, wherein the degree of freedom is horizontal movement of the horizontal beam. It is noted that the horizontal beam has mass m and is assumed to be infinitely stiff, whilst the two vertical beams are assumed massless and with stiffness k.

a) Determine (symbolically) what the undampened angular eigenfrequency ω_u of the equivalent SDOF-system is.

The two beams are connected in parallel, hence:

$$k_{eq} = 2k \implies \omega_u = \sqrt{\frac{2k}{m}}.$$

b) It is noted that (in SI-units) $m = 3$, $k = 30$, $d_0 = -0{,}1$, and $\dot{d}_0 = 0{,}2$. Determine the initial acceleration of the equivalent SDOF-system



From Newton's second law we get:

$$m\ddot{d}(t) + kd(t) = 0 \implies m\ddot{d}_0 + k_{eq}d_0 = 0.$$

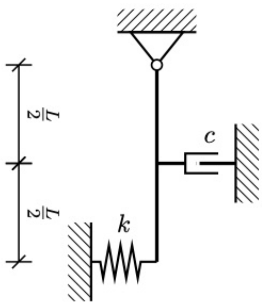
And now we can solve for $\ddot{d}_0$ as:

$$\ddot{d}_0 = \frac{-k_{eq}d_0}{m} = \frac{-60 \cdot (-0{,}1)}{3} = 2\,\frac{\text{m}}{\text{s}}.$$

Lecture 3: Damping of SDOF systems 9. September 2025

3.1 Exercise 3.1

We consider the shown system, which is to be analyzed as an equivalent SDOF system with the rotation of the rod $\theta(t)$ as the degree of freedom. It is assumed that the rod is uniform, infinitely stiff, has a mass of $m = 3\,\text{kg}$, a length $L = 1{,}1\,\text{m}$ and does not dissipate mechanical energy. A linear-viscous damping mechanism, which is producing subcritical damping with $\xi = 0{,}2$ and a spring with stiffness $k = 30\,\frac{\text{N}}{\text{m}}$ is applied as shown. The initial conditions are $\theta_0 = 0$ and $\dot{\theta} = 0{,}1$.



a) Derive the equation of motion for the equivalent SDOF system under the assumption of LTI conditions.

We imagine the rod being displaced a little to the right at the start since $d_0 > 0$. The force the spring will exert is:

$$F_s = -kL\sin\theta.$$

And the force that the damper will exert is

$$F_c = -c\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{L}{2}\sin\theta\right) = c\frac{L}{2}\dot{\theta}\cos\theta.$$

The force from gravity will be:

$$F_g = -mg\frac{L}{2}\sin\theta.$$

Now we take the moments about the support O as:

$$\begin{aligned}
I_O\ddot{\theta} &= F_s + F_c + F_g \\
&= -kL\sin\theta L\cos\theta - c\frac{L}{2}\dot{\theta}\cos\theta\frac{L}{2}\cos\theta - mg\frac{L}{2}\sin\theta \\
&= -kL^2\sin\theta\cos\theta - c\frac{L^2}{4}\dot{\theta}\cos^2\theta - mg\frac{L}{2}\sin\theta.
\end{aligned}$$

Under LTI assumptions we set $\sin\theta \approx \theta$ and $\cos\theta \approx 1$ and get:

$$\begin{aligned}
I_O\ddot{\theta} &= -kL^2\theta - c\frac{L^2}{4}\dot{\theta} - mg\frac{L}{2}\theta \\
&= -\theta\left(kL^2 + mg\frac{L}{2}\right) - c\frac{L^2}{4}\dot{\theta} \\
\implies I_O\ddot{\theta} + \frac{cL^2}{4}\dot{\theta} + \left(kL^2 + mg\frac{L}{2}\right)\theta = 0.
\end{aligned}$$

Now inserting the fact that the moment of inertia of a rod rotating about its end is $I_O = \frac{1}{3}mL^2$ we get:

$$\frac{mL^2}{3}\ddot{\theta} + \frac{cL^2}{4}\dot{\theta} + L\left(kL + \frac{mg}{2}\right)\theta = 0.$$

b) Determine the undamped angular eigenfrequency ω_u.

For translation we normally have that

$$\begin{aligned}
m\ddot{d}(t) + c\dot{d}(t) + kd(t) &= 0 \\
\implies \ddot{d}(t) + \frac{c}{m}\dot{d}(t) + \frac{k}{m}d(t) &= 0 \\
\implies \ddot{d}(t) + 2\left(\omega_d\dot{d}(t) + \omega_u^2 d(t)\right) &= 0.
\end{aligned}$$

An analogous result is applicable for rotation, the eigenfrequency is:

$$\omega_u = \sqrt{\frac{k}{m}} \implies \omega_u = \sqrt{\frac{L\left(kL + \frac{mg}{2}\right)}{\frac{mL^2}{3}}}.$$

This can be simplified as:

$$\begin{aligned}
\omega_u &= \sqrt{\frac{3L\left(kL + \frac{mg}{2}\right)}{mL^2}} \\
&= \sqrt{\frac{3kL + \frac{3mg}{2}}{mL}}.
\end{aligned}$$

c) Solve the equation of motion in a) with the given initial conditions and sketch $\theta(t)$

We have the equation of motion:

$$\frac{mL^2}{3}\ddot{\theta} + \frac{cL^2}{4}\dot{\theta} + L\left(kL + \frac{mg}{2}\right)\theta = 0.$$

Now we set $m_{\text{sub}} \equiv \frac{mL^2}{3}$, $c_{\text{sub}} = \frac{cL^2}{4}$ and $k_{\text{sub}} = L\left(kL + \frac{mg}{2}\right)$. Then

$$m_{\text{sub}}\ddot{\theta} + c_{\text{sub}}\dot{\theta} + k_{\text{sub}}\theta = 0.$$

The solution of this for undercritial damping is:

$$d(t) = e^{-\xi\omega_u t}\left(A_1\cos\left(\omega_d t\right) + A_2\sin\left(\omega_d t\right)\right).$$

Where $\omega_d \equiv \omega_u\sqrt{1-\xi^2}$ and $\xi = \frac{c_{\text{sub}}}{2m_{\text{sub}}\omega_u}$. We can find an expression for ξ as

$$\begin{aligned}\xi &= \frac{c_{\text{sub}}}{2m_{\text{sub}}\omega_u}\\ &= \frac{\frac{cL^2}{4}}{\frac{2mL^2}{3}\omega_u}\\ &= \frac{3c}{8m\omega_u}.\end{aligned}$$

We have $\theta_0 = 0$, $\dot{\theta}_0 = 0{,}1$ so

$$\theta_0 = 0 = A_1.$$

As such we can find:

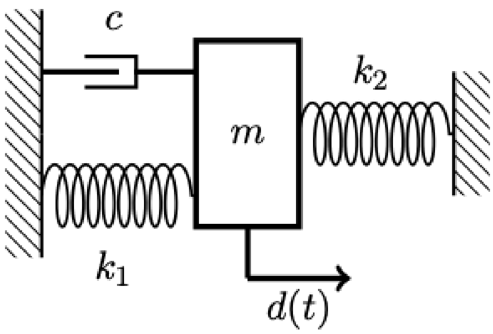
$$\begin{aligned}&\theta(t) = A_2 e^{-\xi\omega_u t}\sin\left(\omega_d t\right)\\ \implies &\dot{\theta}(t) = -\xi\omega_u A_2 e^{-\xi\omega_u t}\sin\left(\omega_d t\right) + \omega_d A_2 e^{-\xi\omega_u t}\cos\left(\omega_d t\right)\\ \implies &\dot{\theta}_0 = \omega_d A_2\\ \implies &A_2 = \frac{\dot{\theta}_0}{\omega_d}.\end{aligned}$$

Therefore the solution is:

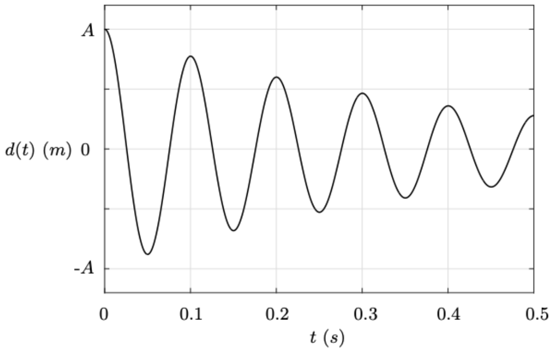
$$\theta(t) = \frac{\dot{\theta}_0}{\omega_d}e^{-\xi\omega_u t}\sin\left(\omega_d t\right).$$

3.2 Exercise 3.2

We consider the shown mass-spring-damper system. Numerical values for the relevant physical parameters will be derived or provided during the exercise. I.e. avoid assuming any numerical values.



I an eigenmotion experiment with $d_0 > 0$ and $\dot{d}_0 = 0$ the movements shown underneath are created in the system. It is noted that the movement amplitude is reduced by 40% after two damped eigenperiods.



a) Determine the damping ratio ξ and determine if the system is undercritically damped, critically damped or overcritically damped.

The dampening ratio δ is defined as

$$\delta \equiv \frac{1}{j} \log\left(\frac{d(t)}{d(t + jT_d)}\right).$$

Where $j \in \mathbb{N}_{>0}$ is chosen such that $t + jT_d$ is the time j damped eigenperipds after t, where each eigenperiod is T_d. In this case we have:

$$\begin{aligned} \delta &= \frac{1}{2} \ln\left(\frac{d(t)}{\frac{3}{5}d(t)}\right) \\ &= \frac{1}{2} \ln \frac{5}{3} \\ &= 0{,}268. \end{aligned}$$

Hence:

$$\xi = \frac{\delta}{\sqrt{\delta^2 + 4\pi^2}} = \frac{0{,}268}{\sqrt{0{,}268^2 + 4\pi^2}} = 0{,}042.$$

Hence the system is undercritically damped.

b) Determine the damped angular eigenfrequency ω_d. (hint: look at the movement graph).

From the plot we see that $T_d = 0{,}1\,\mathrm{s}$, hence:

$$\omega_d = \frac{2\pi}{T_d} = 20\pi\mathrm{s}^{-1}.$$

c) Determine the undamped angular eigenfrequency ω_u.

Here we have:

$$\omega_u = \frac{\omega_d}{\sqrt{1 - \xi^2}} = \frac{20\pi\mathrm{s}^{-1}}{\sqrt{1 - 0{,}042^2}} \approx 62{,}9\,\mathrm{s}^{-1}.$$

d) It is noted that $m = 2{,}2\,\mathrm{kg}$ and $k_1 = 3000\,\frac{\mathrm{N}}{\mathrm{m}}$. Determine k_2.

We know that:

$$k_{\text{eq}} = k_1 + k_2$$

as the springs are connected in parallel. We have that:

$$\omega_u^2 = \frac{k_{\text{eq}}}{m} = \frac{k_1 + k_2}{m} \implies k_2 = \omega_u^2 m - k_1.$$

Plugging in known values we get:

$$k_2 = \left(62{,}9\,\mathrm{s}^{-1}\right)^2 \cdot 2{,}2\,\mathrm{kg} - 3000\,\frac{\mathrm{N}}{\mathrm{m}} = 5{,}7\,\frac{\mathrm{kN}}{\mathrm{m}}.$$

Lecture 4: Tvungen svingning af SDOF-systemer (del I) 16. September 2025

Problem 4.1

Bevis ved brug af "ubestemte koefficienters"-metoden, at den partikulære løsning for et SDOF-system med masse m, dæmpning c og stivhed k, der er påvirket af en harmonisk last med amplitude F_0 og frekvens Ω, er givet ved

$$d_{\mathrm{p}}(t) = \frac{F_0}{\sqrt{\left(k - m\Omega^2\right)^2 + c^2\Omega^2}} \cos\left(\Omega t - \tan^{-1}\left(\frac{c\Omega}{k - m\Omega^2}\right)\right).$$

Derivation

Den generelle formel for tvungen svingning af et SDOF-system er

$$m\ddot{d}(t) + c\dot{d}(t) + kd(t) = F(t).$$

Fra ubestemte koefficienters metode har vi flg. partikulære løsning:

$$d_p(t) = B_1 \cos(\Omega t) + B_2 \sin(\Omega t).$$

Ved indsættelse af den partikulære løsning og den drivende kraft $F = F_0 \cos(\Omega t)$ fås:

$$\begin{aligned}
& m\left(-\Omega^2 B_1 \cos(\Omega t) - \Omega^2 B_2 \sin(\Omega t)\right) \\
& + c\left(-\Omega B_1 \sin(\Omega t) + \Omega B_2 \cos(\Omega t)\right) \\
& + k\left(B_1 \cos(\Omega t) + B_2 \sin(\Omega t)\right) = F_0 \cos(\Omega t) \\
\implies & \left(c\Omega B_2 + kB_1 - \Omega^2 m B_1\right) \cos(\Omega t) \\
& + \left(B_2 k - \Omega c B_1 - \Omega^2 m B_2\right) \sin(\Omega t) = F_0 \cos(\Omega t)
\end{aligned}$$

Vi lader nu $n \in \mathbb{N}_0$, da gælder at:

$$\forall \Omega t = 2n\pi: \qquad \sin(\Omega t) = 0, \quad \cos(\Omega t) = 1.$$

Da får vi ved indsættelse i udtrykket fra før:

$$c\Omega B_2 + \left(k - \Omega^2 m\right) B_1 = F_0. \tag{0.1}$$

Lader vi i stedet $m \in \mathbb{N}_0$, så:

$$\forall \Omega t = 2m\pi + \frac{\pi}{2}: \qquad \sin(\Omega t) = 1, \quad \cos(\Omega t) = 0.$$

Da får vi

$$\begin{aligned}\left(k-\Omega^2 m\right)B_2 - \Omega c B_1 &= 0\\ \implies \quad B_2 &= \frac{\Omega c B_1}{k-\Omega^2 m}\end{aligned} \tag{0.2}$$

Ved indsættelse af Equation (0.2) i Equation (0.1) fås:

$$\begin{aligned}\frac{\Omega^2 c^2 B_1}{k-\Omega^2 m} + \left(k-\Omega^2 m\right)B_1 &= F_0\\ \Omega^2 c^2 B_1 + \left(k-\Omega^2 m\right)^2 B_1 &= F_0\left(k-\Omega^2 m\right)\\ \left(\Omega^2 c^2 + \left(k-\Omega^2 m\right)^2\right)B_1 &= F_0\left(k-\Omega^2 m\right)\\ B_1 &= \frac{F_0\left(k-\Omega^2 m\right)}{\Omega^2 c^2 + \left(k-\Omega^2 m\right)^2}.\end{aligned}$$

Fra Equation (0.2) får vi da at

$$\begin{aligned}B_2 &= \frac{\Omega c F_0\left(k-\Omega^2 m\right)}{\left(k-\Omega^2 m\right)\left(\Omega^2 c^2 + \left(k-\Omega^2 m\right)^2\right)}\\ &= \frac{\Omega c F_0}{\Omega^2 c^2 + \left(k-\Omega^2 m\right)^2}.\end{aligned}$$

Fra definitionen af $B \equiv \sqrt{B_1^2 + B_2^2}$ fås

$$\begin{aligned}B &= \sqrt{\left(\frac{F_0\left(k-\Omega^2 m\right)}{\Omega^2 c^2 + \left(k-\Omega^2 m\right)^2}\right)^2 + \left(\frac{\Omega c F_0}{\Omega^2 c^2 + \left(k-\Omega^2 m\right)^2}\right)^2}\\ &= \sqrt{\frac{F_0^2\left(k-\Omega^2 m\right)^2}{\left(\Omega^2 c^2 + \left(k-\Omega^2 m\right)^2\right)^2} + \frac{\Omega^2 c^2 F_0^2}{\left(\Omega^2 c^2 - \left(k-\Omega^2 m\right)^2\right)^2}}\\ &= \sqrt{\frac{F_0^2\left(k-\Omega^2 m\right)^2 + \Omega^2 c^2 F_0^2}{\left(\Omega^2 c^2 + \left(k-\Omega^2 m\right)^2\right)^2}}\\ &= \sqrt{\frac{F_0^2\left(\Omega^2 c^2 + \left(k-\Omega^2 m\right)\right)}{\left(\Omega^2 c^2 + \left(k-\Omega^2 m\right)^2\right)^2}}\\ &= \frac{F_0}{\sqrt{\Omega^2 c^2 + \left(k-\Omega^2 m\right)^2}}.\end{aligned}$$

Dermed er det vist at den rigtige faktor B er angivet i opgaven. Nu mangler vi blot at vise at fasevinklen $\epsilon = \tan^{-1}\left(c\Omega/\left(k-m\Omega^2\right)\right)$ også er korrekt. Denne er defineret som $\epsilon = \tan^{-1}\left(B_2/B_1\right)$. Ved indsættelse fås:

$$\begin{aligned}\epsilon &= \tan^{-1}\left(\frac{\Omega c F_0}{F_0\left(k-\Omega^2 m\right)}\right)\\ &= \tan^{-1}\left(\frac{\Omega c}{k-\Omega^2 m}\right).\end{aligned}$$

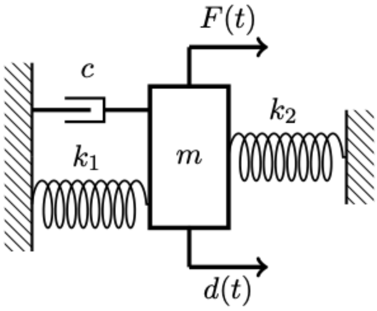
Dermed er det vist.

$$\boxed{\begin{aligned}B &= \frac{F_0}{\sqrt{\Omega^2 c^2 + \left(k-\Omega^2 m\right)^2}}\\ \epsilon &= \tan^{-1}\left(\frac{\Omega c}{k-\Omega^2 m}\right).\end{aligned}}$$

RESULT

Problem 4.2

Vi returnerer til systemet behandlet i opgave 3.2 (se forrige lektion), som nu påføres en harmonisk last. Denne last er givet ved $F(t) = F_0 \cos(\Omega t)$, hvor $F_0 = 1\,\text{kN}$, mens Ω ønskes bestemt. I denne forbindelse oplyses det, at fasevinklen i den stationære svingning er lig 90°. a) Bestem lastfrekvensen, Ω. b) Bestem flytningsamplituden i den stationære svingning, B. c) Bestem den dynamiske forstærkningsfaktor, D.



Derivation

Eftersom Fasevinklen i den stationære svingning er 90° har vi $\tan 90° \to \infty \implies p = 1 \implies \Omega = \omega_u = 62{,}9\,\text{s}^{-1}$.

Eftersom $p = 1$ har vi

$$\begin{aligned} D \equiv \frac{Bk}{F_0} &= \frac{1}{\sqrt{\left(1-p^2\right)^2 + 4\xi^2 p^2}} \\ &= \frac{1}{\sqrt{4\xi^2}} \\ &= \frac{1}{2\xi} \\ \implies \qquad B &= \frac{F_0}{2\xi k_{\text{eq}}}. \end{aligned}$$

Hvor $k_{\text{eq}} = k_1 + k_2$ og ξ er fundet/givet i opgave 3.2. Dermed er $D = \frac{1}{2\xi}$ også fundet.

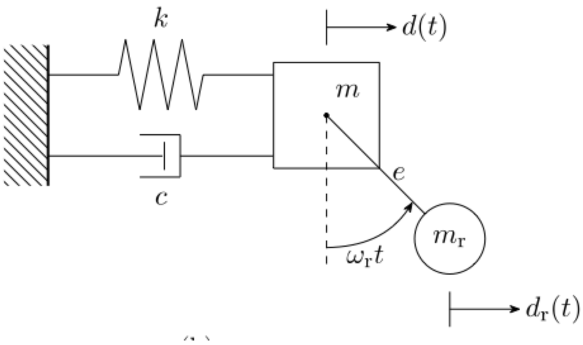
$$\boxed{\begin{aligned} &\text{a)} \quad \Omega = \omega_u = 62{,}9\,\text{s}^{-1} \\ &\text{b)} \quad B = \frac{F_0}{2\xi k_{\text{eq}}} \\ &\text{c)} \quad D = \frac{1}{2\xi}. \end{aligned}}$$

RESULT

Problem 4.3

Bevis, at den styrende bevægelsesligning for SDOF-systemet skitseret på figur 3.6b i forelæsningsnoterne kan formuleres som

$$(m + m_r)\,\ddot{d}(t) + c\dot{d}(t) + kd(t) = m_r\omega_r^2 e\sin(\omega_r t)\,.$$



Figur 0.3: SDOF-systemet fra figur 3.6b i forelæsningen.

Derivation

Fra Newtons anden lov fås

$$m\ddot{d}(t) + m_r\ddot{d}_r(t) = -c\dot{d}(t) - kd(t).$$

$d_r(t)$ kan også skrives som

$$d_r(t) = d(t) + e\sin(\omega_r t)\,.$$

Dermed har vi

$$\ddot{d}_r(t) = \ddot{d}(t) - \omega_r^2 e\sin(\omega_r t)\,.$$

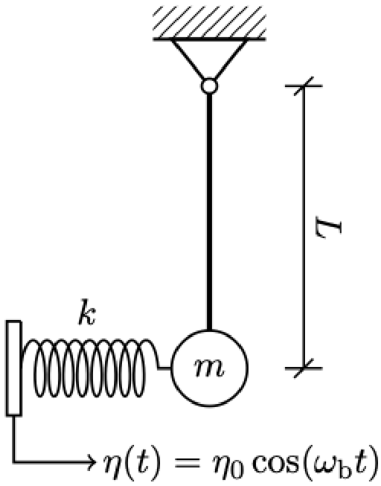
Ved indsættelse af dette i Newtons 2. lov får vi

$$\begin{aligned} m\ddot{d}(t) + m_r\left(\ddot{d}(t) - \omega_r^2 e\sin(\omega_r t)\right) + c\dot{d}(t) + kd(t) &= 0 \\ (m + m_r)\,\ddot{d}(t) + c\dot{d}(t) + kd(t) &= m_r\omega_r^2 e\sin(\omega_r t)\,. \end{aligned}$$

$$\boxed{(m + m_r)\ddot{d}(t) + c\dot{d}(t) + kd(t) = m_r\omega_r^2 e\sin(\omega_r t)}$$ RESULT

Problem 4.4

Vi betragter det skitserede pendulsystem og antager, at stangen er uendelig stiv, uniform, masseløs og uden dæmpning. En bevægelig understøtning virker, som skitseret, med $\eta(t) = \eta_0 \cos(\omega_b t)$. Systemet skal modelleres som et SDOF-system med frihedsgraden værende rotation af stangen, $\theta(t)$. a) Udled bevægelsesligningen for $\theta(t)$ under antagelsen om LTI-betingelser. b) Bestem (symbolsk) den udæmpede vinkelegenfrekvens, ω_u.



Derivation

Momentet omkring stangen (idet den roteres en lille vinkel θ væk fra fjederen) giver

$$I\ddot{\theta} = -k\left(L\sin\theta - \eta\right)L\cos\theta - mgL\sin\theta.$$

Vi bruger $\cos\theta \approx 1$ og $\sin\theta \approx \theta$ og får

$$I\ddot{\theta} = -k\left(L\theta - \eta\right)L - mgL\theta.$$

$\eta(t) = \eta_0 \cos(\omega_b t)$ er givet i opgaven, hvorfor

$$\begin{aligned} I\ddot{\theta} &= -k\left(L\theta - \eta_0\cos(\omega_b t)\right)L - mgL\theta \\ I\ddot{\theta} + \left(kL^2 + mgL\right)\theta &= kL\eta_0\cos(\omega_b t). \end{aligned}$$

For en stiv stang der roterer om enden er $I = mL^2$ og vi får derfor

$$mL^2\ddot{\theta} + \left(kL^2 + mgL\right)\theta = kL\eta_0\cos(\omega_b t).$$

For at bestemme den udæmpede vinkelegenfrekvens bruger vi blot

$$\begin{aligned} \omega_u &= \sqrt{\frac{B_2}{B_1}} \\ &= \sqrt{\frac{kL^2 + mgL}{mL^2}} \\ &= \sqrt{\frac{kL + mg}{mL}}. \end{aligned}$$

$$\boxed{\begin{aligned} &\text{a)} \quad mL^2\ddot{\theta} + \left(kL^2 + mgL\right)\theta = kL\eta_0\cos(\omega_b t) \\ &\text{b)} \quad \omega_u = \sqrt{\frac{kL + mg}{mL}}. \end{aligned}}$$

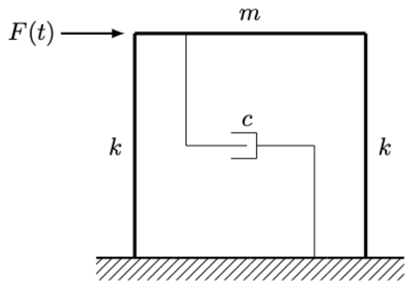
RESULT

Lecture 5: Tvungen svingning af SDOF-systemer (del II)

23. September 2025

Problem 5.1

Vi betragter den viste rammestruktur, der er påført begyndelsesbetingelserne $d_0 \neq 0$ og $\dot{d}_0 \neq 0$ og er belastet af en enhedssteplast, som aktiveres ved $t_1 > 0$. Det oplyses, at den tværgående bjælke har massen m og er uendelig stiv, mens hver søjle har stivheden k og er masseløs. Rammestrukturen ønskes beskrevet som et SDOF-system, hvori den horisontale flytning af tværbjælken er frihedsgraden. Dæmpningsmekanismen yder underkritisk dæmpning. a) Bestem den ækvivalente masse, ækvivalente stivhed og ækvivalente dæmpning. b) Bestem systemets flytningsrespons. c) Bestem det stationære flytningsrespons. d) Bestem det stationære hastighedsrespons.



Derivation

Vi kan betragte systemet som en masse påvirket af en last, holdt fast af to parallelforbundne fjedre og med dæmpning i både positiv og negativ bevægelsesretning. Dæmpningnen ovenfor er blot $c_{\text{eq}} = c$, de parallelforbundne fjedre har en ækvivalent fjederkonstant på $k_{\text{eq}} = 2k$ og en ækvivalent masse på m.

Vi har den generelle løsning

$$d(t) = d_{\text{h}}(t) + d_{\text{p}}(t).$$

Idet der er underkritisk dæmpning har vi $\xi \in (0,1)$. Hvilket giver den homogene løsning

$$d_{\text{h}}(t) = Ae^{-\xi\omega_{\text{u}}t}\cos\left(\omega_{\text{d}}t - \psi\right).$$

Vha. Duhamels integrale fås

$$\begin{aligned} d_{\text{p}}(t) &= \int_{t_1}^{t} H(t-\tau)F(\tau)\,\mathrm{d}\tau \\ &= \int_{t_1}^{t} \frac{1}{m\omega_{\text{d}}} e^{-\xi\omega_{\text{u}}(t-\tau)} \sin\left(\omega_{\text{d}}\left(t-\tau\right)\right) \cdot 1\,\mathrm{d}\tau \\ &= \frac{1}{m\omega_{\text{d}}} e^{-\xi\omega_{\text{u}}t} \int_{t_1}^{t} e^{\xi\omega_{u}\tau} \sin\left(\omega_{\text{d}}\left(t-\tau\right)\right)\mathrm{d}\tau. \end{aligned}$$

Vi kan nu løse integralet som

$$\begin{aligned} \int_{t_1}^{t} e^{\xi\omega_{u}\tau} \sin\left(\omega_{d}\left(t-\tau\right)\right)\mathrm{d}\tau &= \frac{1}{\omega_{\text{d}}^2 + \xi^2\omega_{\text{u}}^2}\left[e^{\xi\omega_{\text{u}}\tau}\left(\omega_{\text{d}}\cos\left(\omega_{\text{d}}\left(t-\tau\right)\right) + \xi\omega_{\text{u}}\sin\left(\omega_{\text{d}}\left(t-\tau\right)\right)\right)\right]_{t_1}^{t} \\ &= \frac{1}{\omega_{\text{u}}^2}\left(\omega_{\text{d}}e^{\xi\omega_{\text{u}}t} - \omega_{\text{d}}e^{\xi\omega_{\text{u}}t_1}\cos\left(\omega_{\text{d}}\left(t-t_1\right)\right) - \xi\omega_{\text{u}}e^{\xi\omega_{\text{u}}t_1}\sin\left(\omega_{\text{d}}\left(t-t_1\right)\right)\right) \\ &= \frac{1}{\omega_{\text{u}}^2}\left(\omega_{\text{d}}e^{\xi\omega_{\text{u}}t} - Be^{\xi\omega_{\text{u}}t_1}\cos\left(\omega_{\text{d}}\left(t-t_1\right)-\epsilon\right)\right). \end{aligned}$$

Hvor $B = \sqrt{\omega_{\text{d}}^2 + \xi^2\omega_{\text{u}}^2} = \omega_{\text{u}}$ og $\epsilon = \tan^{-1}\left(\xi/\sqrt{1-\xi^2}\right)$.

Ved indsættelse af dette i Duhamels integrale fås:

$$\begin{aligned} d_{\text{p}}(t) &= \frac{e^{-\xi\omega_{\text{u}}t}}{m\omega_{\text{d}}\omega_{\text{u}}^2} \cdot \left(\omega_{\text{d}}e^{\xi\omega_{\text{u}}t} - e^{\xi\omega_{\text{u}}t_1}\cos\left(\omega_{\text{d}}\left(t-t_1\right)-\epsilon\right)\right) \\ &= \frac{\omega_{\text{d}}}{m\omega_{\text{d}}\omega_{\text{u}}^2} - \frac{e^{-\omega_{\text{u}}\xi(t-t_1)}}{m\omega_{\text{d}}\omega_{\text{u}}}\cos\left(\omega_{\text{d}}\left(t-t_1\right)-\epsilon\right) \end{aligned}$$

$$= \frac{1}{k_{\text{eq}}} - \frac{1}{m\omega_{\text{u}}^2\sqrt{1-\xi^2}} e^{-\xi\omega_{\text{u}}(t-t_1)} \cos\left(\omega_{\text{d}}\left(t-t_1\right)-\epsilon\right)$$
$$= \frac{1}{k_{\text{eq}}} - \frac{1}{k_{\text{eq}}\sqrt{1-\xi^2}} e^{-\xi\omega_{\text{u}}(t-t_1)} \cos\left(\omega_{\text{d}}\left(t-t_1\right)-\epsilon\right).$$

Dermed bliver vores totale løsning:

$$d(t) = Ae^{-\xi\omega_{\text{u}}t}\cos\left(\omega_{\text{d}}t-\psi\right) + \frac{1}{k_{\text{eq}}} - \frac{1}{k_{\text{eq}}\sqrt{1-\xi^2}} e^{-\xi\omega_{\text{u}}(t-t_1)} \cos\left(\omega_{\text{d}}\left(t-t_1\right)-\epsilon\right).$$

For $t \to \infty$ ses at $d \to \frac{1}{k_{\text{eq}}}$ hvilket derfor er den stationære flytnignsrespons.

For $t \to \infty$ fås $\dot{d} \to 0$ hvilket da er den stationære hastighedsrespons.

a) $c_{\text{eq}} = c, m_{\text{eq}} = m, k_{\text{eq}} = 2k$

b) $d(t) = Ae^{-\xi\omega_{\text{u}}t}\cos\left(\omega_{\text{d}}t-\psi\right) + \frac{1}{k_{\text{eq}}} - \frac{1}{k_{\text{eq}}\sqrt{1-\xi^2}} e^{-\xi\omega_{\text{u}}(t-t_1)} \cos\left(\omega_{\text{d}}\left(t-t_1\right)-\epsilon\right)$

c) $\frac{1}{k_{\text{eq}}}$

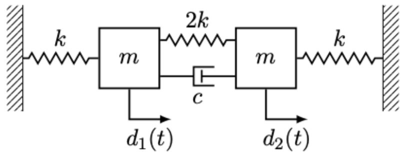
d) 0.

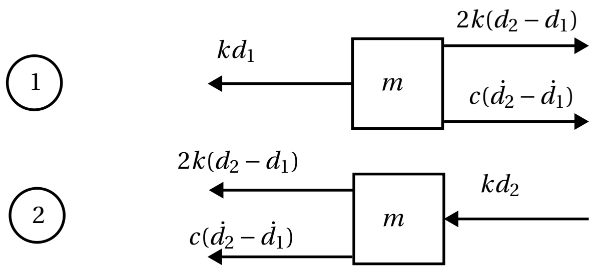
RESULT

Lecture 7: Fri svingning/egensvingning af MDOF-systemer (del II) 28. Oktober 2025

Problem 7.1

Vi returnerer til systemet behandlet i opgave 6.1 og tilføjer en dæmpningsmekanisme som vist. Bevis, at dæmpningen er klassisk.





Figur 0.4: Free body diagram of the system

Derivation

Baseret på Figure 0.4 kan Newtons 2. lov anvendes som

$$
\begin{aligned}
m \cdot \ddot{d}_1 &= 2kd_2 - 3kd_1 + c(\dot{d}_2 - \dot{d}_1) \\
m \cdot \ddot{d}_1 + 3kd_1 - 2kd_2 + c\dot{d}_1 - c\dot{d}_2 &= 0 \\
m \cdot \ddot{d}_2 &= -3kd_2 + 2kd_1 - c(\dot{d}_2 - \dot{d}_1) \\
m \cdot \ddot{d}_2 - 2kd_1 + 3kd_2 + c\dot{d}_2 - c\dot{d}_1 &= 0.
\end{aligned}
$$

Dermed har vi

$$
M = \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix}, \quad C = \begin{bmatrix} c & -c \\ -c & c \end{bmatrix}, \quad K = \begin{bmatrix} 3k & -2k \\ -2k & 3k \end{bmatrix}.
$$

Fra Opgave 6.1 har vi at $\Phi = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ Desuden har vi $\tilde{C}$ givet som

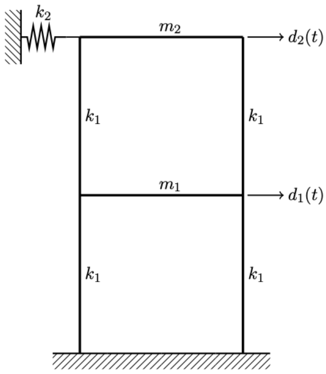
$$
\tilde{C} \equiv \Phi^{\mathsf{T}} C \Phi.
$$

Dermed er $\tilde{C}$:

$$
\tilde{C} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}^{\mathsf{T}} \begin{bmatrix} c & -c \\ -c & c \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 4c \end{bmatrix}.
$$

Eftersom $\tilde{C} = \begin{bmatrix} 0 & 0 \\ 0 & 4c \end{bmatrix}$ er diagonal må der derfor være tale om klassisk dæmpning.

Problem 7.2



Vi betragter det viste rammesystem. Heri gælder det, at de horisontale bjælker er uendeligt stive og har masserne m_1 og m_2, mens de vertikale bjælker (også kaldet søjler) er fleksible (alle med stivheden k_1) og masseløse. En fjeder med stivheden k_2 er påsat den øverste horisontale bjælke. Systemet modelleres som et 2DOF-system, hvori den første frihedsgrad, $d_1(t)$, er den horisontale flytning af den nederste horisontale bjælke, mens den anden frihedsgrad, $d_2(t)$, er den horisontale flytning af den øverste horisontale bjælke. Det oplyses, at de udæmpede vinkelegenfrekvenser og de tilhørende masse-normaliserede egenvektorer af 2DOF-systemet er

$$\omega_1 = \sqrt{200}\,\text{rad/s}, \qquad \omega_2 = \sqrt{600}\,\text{rad/s}, \qquad \phi_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \phi_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

a) Bestem værdierne for k_1 og k_2.
b) Bestem værdierne for m_1 og m_2.
Hvad angår 2DOF-systemet, så oplyses begyndelsesbetingelserne $d_0 = \begin{pmatrix} 0{,}1 & 0{,}2 \end{pmatrix}^\mathsf{T}$ og $\dot{d}_0 = \begin{pmatrix} 0 & 0 \end{pmatrix}^\mathsf{T}$. Endvidere oplyses det, at systemet er klassisk dæmpet med de modale dæmpningskonstanter $\tilde{c}_{11} = 8$ og $\tilde{c}_{22} = 10$.
c) Bestem dæmpningsmatricen, $C \in \mathbb{R}^{2\times 2}$.
d) Bestem begyndelsesaccelerationsvektoren, $\ddot{d}_0$.

Derivation

Vi har fået givet to egenfrekvenser og de dertilhørende egenvektorer. Vi har altså:

$$\Lambda = \begin{bmatrix} 200\,\text{s}^{-1} & 0 \\ 0 & 600\,\text{s}^{-1} \end{bmatrix}, \qquad \Phi = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad \tilde{M} = I.$$

Fra undervisningen har vi at siden vi generelt har

$$\Phi^\mathsf{T} K \Phi = \Phi^\mathsf{T} M \Phi \Lambda$$

så medfører $\tilde{M} = I$ at $\tilde{K} = \Lambda$. $\tilde{K}$ er givet som:

$$\tilde{K} \equiv \Phi^\mathsf{T} K \Phi.$$

Altså:

$$\begin{aligned} \tilde{K} &= \Phi^\mathsf{T} K \Phi \\ K &= \left(\Phi^\mathsf{T}\right)^{-1} \tilde{K} \Phi^{-1} \\ &= \begin{bmatrix} 200 & -100 \\ -100 & 200 \end{bmatrix}. \end{aligned}$$

Vi har

$$K = \begin{bmatrix} 4k_1 & -2k_1 \\ -2k_1 & 2k_1 + k_2 \end{bmatrix} \implies k_1 = 50,\ k_2 = 100.$$

Samme procedure kan følges for masserne. Siden $\tilde{M} = I$ fås:

$$\begin{aligned}\tilde{M} &= \Phi^{\mathsf{T}} M \Phi \\ M &= \left(\Phi^{\mathsf{T}}\right)^{-1} \Phi \\ &= \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}.\end{aligned}$$

Dermed har vi $m_1 = m_2 = \frac{1}{2}$.

Udnyttes samme teknik for dæmpningsmatricen får vi

$$\begin{aligned}\tilde{C} &= \Phi^{\mathsf{T}} C \Phi \\ C &= \left(\Phi^{\mathsf{T}}\right)^{-1} \tilde{C} \Phi^{-1} \\ &= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 8 & 0 \\ 0 & 10 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} 9 & -1 \\ -1 & 9 \end{bmatrix}.\end{aligned}$$

For at bestemme $\ddot{d}_0$ benytter vi den generelle formel

$$\begin{aligned}M\ddot{d}_0 + C\dot{d}_0 + Kd_0 = 0 \\ \ddot{d}_0 &= -M^{-1}\left(C\dot{d}_0 + Kd_0\right) \\ &= -\begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}^{-1} \left(\frac{1}{2}\begin{bmatrix} 9 & -1 \\ -1 & 9 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 200 & -100 \\ -100 & 200 \end{bmatrix} \cdot \begin{bmatrix} 0{,}1 \\ 0{,}2 \end{bmatrix}\right) \\ &= \begin{bmatrix} 0 \\ -60 \end{bmatrix}.\end{aligned}$$

a) $k_1 = 50,\ k_2 = 100$

b) $m_1 = m_2 = \dfrac{1}{2}$

c) $C = \dfrac{1}{2}\begin{bmatrix} 9 & -1 \\ -1 & 9 \end{bmatrix}$

d) $\ddot{d}_0 = \begin{bmatrix} 0 \\ -60 \end{bmatrix}.$

RESULT

Lecture 8: Fri svingning/egensvingning af MDOF-systemer (del III)

4. November 2025

Problem 8.1

Vi returnerer til systemet behandlet i Opgave 7.1. Bevis, at dæmpningsforholdene er givet som

$$\xi_1 = 0, \quad \xi_2 = \frac{c}{m\omega_2}.$$

Derivation

Fra forelæsningen har vi at $\tilde{C}$ er givet som

$$\tilde{C}_{jj} = 2\xi_j \tilde{m}_{jj}\omega_j.$$

Heri kan dæmpningsforholdet ξ_j isoleres som

$$\xi_j = \frac{\tilde{C}_{jj}}{2\tilde{M}_{jj}\omega_j}.$$

Hvori $\tilde{C}$ er fundet i opgave 7.1 til

$$\tilde{C} = \begin{bmatrix} 0 & 0 \\ 0 & 4c \end{bmatrix}.$$

Fra Opgave 6.1 har vi hhv:

$$\omega_1 = \sqrt{\frac{k}{m}}, \quad \omega_2 = \sqrt{5}\omega_1.$$

De modale masser bestemmes som

$$\begin{aligned}
\tilde{M} &= \Phi^\mathsf{T} M \Phi \\
&= \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \\
&= 2m \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.
\end{aligned}$$

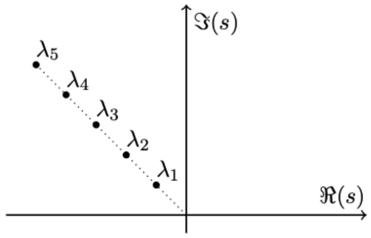
Dermed har vi at $\tilde{M}_{11} = \tilde{M}_{22} = 2m$, hvorfor vi får:

$$\boxed{\begin{aligned}
\xi_1 &= \frac{0}{2\cdot 2m\cdot\omega_1} = 0 \\
\xi_2 &= \frac{4c}{2\cdot 2m\cdot\omega_2} = \frac{c}{m\omega_2}.
\end{aligned}}$$

RESULT

Problem 8.2

Et eksperiment har–under antagelsen om klassisk dæmpning–resulteret i de nedenfor viste egenværdier (og deres komplekst konjugerede), der ligger på en ret linje fra origo. Det oplyses, at $\lambda_1 = -0{,}900 + 8{,}96i$ og $\lambda_2 = -2{,}63 + 26{,}1i$. (a) Bestem de fem dæmpningsforhold.



(b) Bestem de to første udæmpede vinkelegenfrekvenser.
Det oplyses nu, at

$$K = 1000\begin{bmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 1 \end{bmatrix}, \qquad M = I.$$

(c) Bestem dæmpningsmatricen, $C \in \mathbb{R}^{5\times 5}$, ved brug af Rayleighs dæmpningsmodel baseret på de to første egensvingningskonfigurationer.
(d) Hvor store fejl yder Rayleighs dæmpningsmodel på ξ_3, ξ_4 og ξ_5?

Derivation

Fra Forelæsning 7 har vi at der for klassisk dæmpning gælder at:

$$\{\lambda_j, \lambda_{j+n}\} = -\xi_j\omega_j \pm i\omega_j\sqrt{1-\xi_j^2}, \text{ hvor } \xi_j \equiv \frac{\tilde{C}_{jj}}{2\tilde{M}_{jj}\omega_j} \text{ for } \omega_j > 0.$$

Se svaret i svararket for resten af den her trælse opgave...